Problem set #3

Due on November 25th by 12.30pm. Send solution to dealberti.francesco@gmail.com
Answers must be typed. You can work in teams under the constraint: Max(Team Size)=2

1. (**Reweighting**) Go to David Autor's webpage and download the cleaned 1979 and 1997 MORG files from:

   https://economics.mit.edu/people/faculty/david-h-autor/data-archive

   the paper is Trends in U.S. Wage Inequality: Revising the Revisionists.

   a) Read through the cleaning programs associated with the MORG files. Describe the important decisions being made.

   b) Make a new variable "lnhr_wage" which is the log of the "hr_wage" variable. Restrict attention to the cleaned sample with "hr_wage_sample==1". Use the "kdensity" command to plot kernel density estimates of lnhr_wage using the variable wgt_hrs (which is the product of the sample weights and the number of hours worked) as an "aweight". Make density plots for both years on the same grid.

   c) How has the wage distribution changed over time?

   d) A number of demographic shifts occured between 1979 and 1997 that might alter the aggregate wage distribution even if the distribution for each demographic subgroup remained unaltered. Repeat the exercise in b) separately for black men age 25-50 and for white women age 25-50. How are the patterns different?

   e) One way of adjusting the distribution for compositional differences is to use propensity score reweighting.

   Append the 1979 and 1997 files together and create a dummy variable equal to one for observations in the 1997 sample. Estimate a logit of the 1997 dummy on a dummy for female, a black dummy, an "other race" dummy, and a quadratic polynomial in age. Use the "aweights" when estimating the logit.

   f) Use the propensity score implied by this logit to reweight the 1997 wage distribution so that it has the 1979 demographic distribution. Make a kernel density plot of the actual 1979 and 1997 wage distributions against the "counterfactual" reweighted 1997 distribution.

   g) Use the propensity score to reweight the 1979 wage distribution so that it has the 1997 demographic distribution (if you get stuck read the Dinardo, Fortin, and Lemieux paper). Make a kernel density plot of the actual 1979 and 1997 wage distributions against the "counterfactual" reweighted 1979 distribution. Be sure to use the product of the wgt_hrs variable and the propensity score based weights when computing the kernel density.

   h) Based upon your analysis what would you say was the effect of the demographic changes between 1979 and 1997 on the evolution of the aggregate wage distribution?

   i) Examine the suitability of your propensity score specification by comparing the reweighted mean, standard deviation, and the 10th, 25th, 50th, 75th, and 90th percentiles of age in the 1997 sample to the equivalent moments in the 1979 sample.

j) Do the same exercise restricting attention to blacks.

k) Try experimenting with different specifications for age in the propensity score logit and checking how the results of i) and j) change.

l) What do you conclude about the suitability of the propensity score specification?

2. (**Duflo, Hanna and Ryan (2012)**) This question asks you to solve and estimate a simplified version of the model from Duflo, Hanna and Ryan (2012). Consider a teacher deciding between work and leisure in two time periods, $t \in \{0, 1\}$. Let $L_t \in \{0, 1\}$ indicate whether the teacher takes leisure in period t. The teacher's flow utility in period t is given by

$$U_t = \beta C_t + L_t \cdot (\mu + \varepsilon_t),$$

where C_t is consumption in period t. All income and consumption is assumed to happen in period 2, so $C_1 = 0$. In period 2, the teacher receives a salary payment of 5, and an extra bonus of 5 if she worked in both periods (that is, if $L_1 = L_2 = 0$). Assume that the teacher has a discount factor $\delta \in [0, 1]$.

   (a) What are the state variables in period 2? Write an expression for the teacher's value function in period 2.

   (b) Suppose that $\varepsilon_t \sim \mathcal{N}(0, 1)$. Compute the teacher's expected second-period payoff. Use this to write an expression for the teacher's value function in period 1.

   (c) Suppose you have a sample of teachers' choices (L_{1i}, L_{2i}) for $i \in \{1, \dots, N\}$. Write an expression for the likelihood of teacher i's choice sequence, $\Pr[(L_1 = L_{1i}) \cap (L_2 = L_{2i})]$, as a function of the parameters β, μ and δ. Use this to write an expression for the sample log-likelihood.

   (d) Download the data set `‘‘dhr_example.csv"` from the course website. This data set includes 800 observations on teachers' choice sequences (L_{1i}, L_{2i}). Import the data into a statistical software package. Estimate the parameters (β, μ, δ) by maximum likelihood, and report your parameter estimates.

   (e) Estimate an alternative version of the model where you assume that teachers are not forward-looking. Use your results from the two models to test the null hypothesis that teachers are not forward-looking. Discuss intuitively the features of the data that allow you to determine whether teachers are forward-looking.

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   (g) Estimate an alternative version of the model where $\mu \sim N(\mu_1; \sigma_1)$. How does this version of the model differs from the previous one? Write down the associated likelihood and provide estimate of $(\beta, \delta, \mu_1, \sigma_1,)$ along with their standard errors using SML. Make sure to explain how you derived the standard errors for your coefficients.